

Controllable Block

Declare which labels the controller is allowed to restrict:

```
automaton Actuator {
  controllable {
    label extend;
    label retract;
  }
  // ...
}
```

- Only **controllable** labels can be disabled by the synthesized controller.
- Every label *not* listed in **controllable** is **uncontrollable** by default.

Uncontrollable Labels

Uncontrollable labels represent environment events:

- Sensor readings, faults, external requests.
- The controller **cannot prevent** them — they always pass through.
- The controller must satisfy the property *regardless* of when uncontrollable events occur.

Controller Synthesis (DSL)

```
controller SafeCtrl {
  source Plant;
  satisfying Safety;

  // Optional directives:
  minimize;
  export;
  diagnostics {
    counterexample;
    deadlock_traces;
    max_counter_traces 5;
    proof_obligations;
  }
}
```

Directive	Effect
source	Automaton/composition to control
satisfying	Formula the controller must enforce
minimize	Reduce controller states
export	Write controller CLTS to file
diagnostics	Enable debug output

CLI

```
mununu context synth file.ctxdsl \  
  --formula Safety \  
  --automaton Plant \  
  --minimize \  
  --counterexample
```

Synthesis Results

Outcome	Meaning
Realizable	A controller exists; states count reported.
Unrealizable	No controller can enforce the property under all uncontrollable behaviors.

When realizable, the output includes:

- Number of controller states.
- Winning state set (states from which the property can be enforced).
- Optionally: minimized controller, exported CLTS.

When unrealizable, enable **diagnostics** to obtain counterexample traces showing how the environment can force a violation.

Game-Theoretic Formulas

Game-theoretic reachability:

| $\mu X. (\text{target} \mid \mid [(\text{ctrl} = \text{controllable})] X)$

Computes the **controllable attractor** — states where the controller can force reaching **target**.

Important Rules

1. **No overlap:** controllable labels must not appear as uncontrollable in multiple automata within the same composition.
2. **Restriction only:** the controller can only *disable* controllable actions — it cannot force them to occur.
3. **Pass-through:** uncontrollable events always pass through; the controller must tolerate every possible sequence of environment actions.

Further Reading

Ramadge & Wonham (1987) “Supervisory Control of a Class of Discrete Event Processes” — controllability theory. Arnold, Vincent & Walukiewicz (2003) — games for mu-calculus synthesis.